

Week 11 – Final Project

Introduction

This final project will show the development of an end to end big data pipeline using technologies from Hadoop. It shows the usage of different programs like Nifi for data ingestion, HDFS for distributed storage, Hive for data warehousing, Spark for distributed processing and machine learning, and HBase for NoSQL storage of model results. Throughout the project, I will be analyzing a Vehicle Dataset and applying machine learning models to predict selling prices based on vehicle features.

Dataset

Vehicle Sales Data

<https://www.kaggle.com/datasets/syedanwarafri/vehicle-sales-data>
carprices.csv

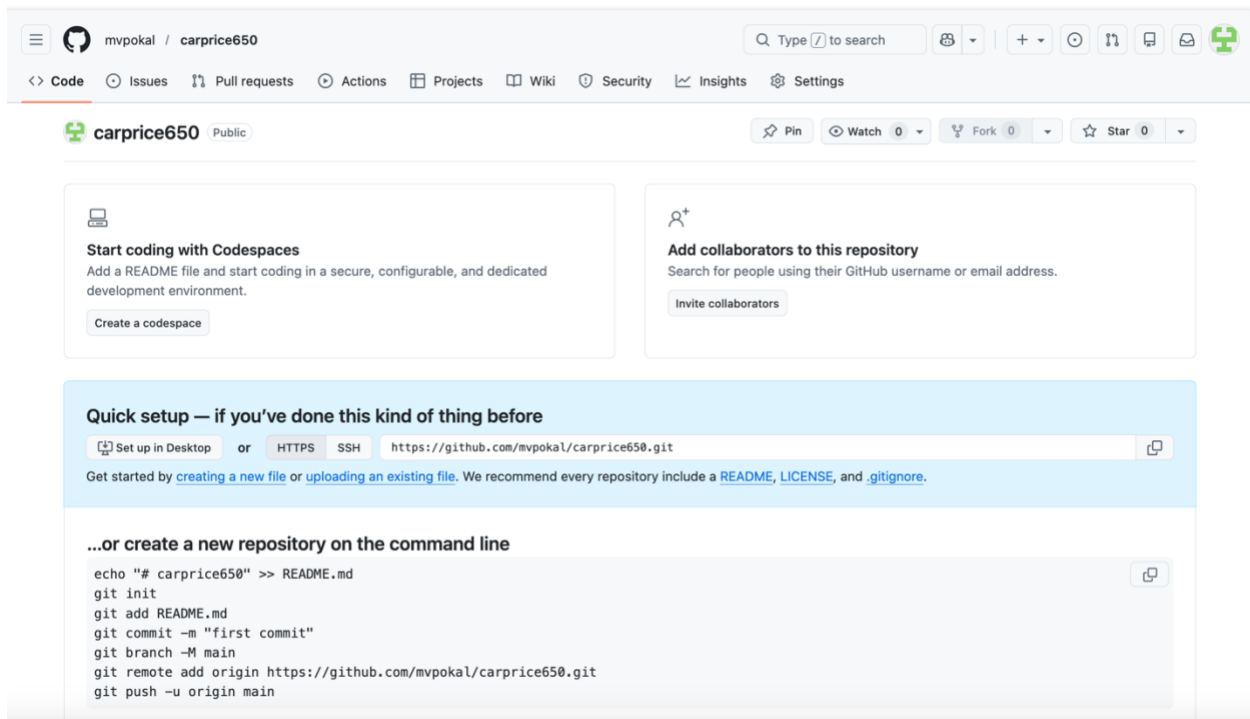
This data set had 16 columns and about 65,000 rows, so I had to delete a lot of rows for this assignment. I condensed it to 61 rows and only looked at the year 2015. To make sure that the VM pipeline could run successfully.

Pipeline Overview

I will be using six major technologies. Nifi to download the dataset from GitHub. HDFS helps store the dataset in the distributed file system. YARN manages the cluster resources for Spark. Hive creates a structured warehouse table. Spark will perform distributed processing and machine learning. HBase will store machine learning evaluation metrics.

Objective 1 – NiFi Data Ingestion into HDFS:

Created a new GitHub repository named carprice650.
<https://github.com/mvpokal/carprice650.git>

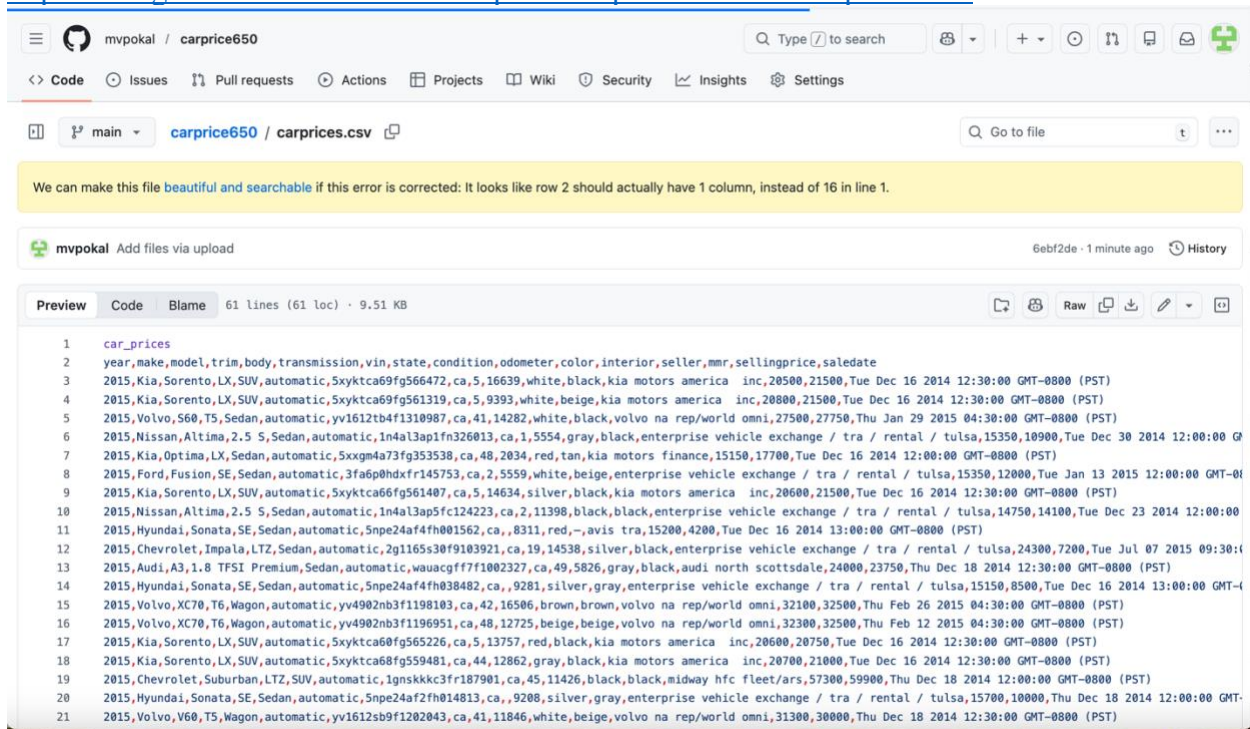


Carprices.csv has been uploaded into the new Github rep, carprice650.

Raw Github link:

<https://raw.githubusercontent.com/mvpokal/carprice650/refs/heads/main/carprices.csv>

<https://raw.githubusercontent.com/mvpokal/carprice650/main/carprices.csv>



ssh -L 9870:localhost:9870 -L 8088:localhost:8088 -L 8080:localhost:8080 -L 18080:localhost:18080 -L 16010:localhost:16010 -L 8983:localhost:8983 -L 8443:localhost:8443 meganpokal27@35.236.15.189

Logged into hadoop-hive-spark-hbase in the terminal.

```
meganpokal — meganpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-...
...p-hive-spark-hbase — -bash      ...anpokal27@35.236.15.189      ...pokal27@35.236.15.189      ... +
Last login: Wed Feb 25 04:15:55 2026 from 174.233.226.84
[meganpokal27@dsc650bigdata:~$ ls
dsc650-infra
[meganpokal27@dsc650bigdata:~$ cd dsc650-infra/bellevue-bigdata
[meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata$ ls
hadoop-hive-spark-hbase  kafka  nifi  solr
[meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata$ cd hadoop-hive-spark-hbase
[meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$ docker-compose up
p -d
Creating network "hadoop-hive-spark-hbase_spark_net" with the default driver
Creating hadoop-hive-spark-hbase_zoo1_1      ... done
Creating hadoop-hive-spark-hbase_zoo3_1      ... done
Creating hadoop-hive-spark-hbase_zoo2_1      ... done
Creating hadoop-hive-spark-hbase_hivemetastore_1 ... done
Creating hadoop-hive-spark-hbase_worker2_1   ... done
Creating hadoop-hive-spark-hbase_worker1_1   ... done
Creating hadoop-hive-spark-hbase_master_1    ... done
[meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS
PORTS
NAME

S
fcbf8dd7e8c8   nasheb/bigdata:1.0                 "/entrypoint.sh"        11 seconds ago Up 8 seconds (health:
starting)      1-8041/tcp, 8043-8981/tcp, 0.0.0.0:8982->8982/tcp, [::]:8982->8982/tcp, 8983-9864/tcp, 9
866-45079/tcp, 45081-65535/tcp, 0.0.0.0:9865->9865/tcp, [::]:9865->9865/tcp, 0.0.0.0:8102->8042/tcp,
[::]:8102->8042/tcp, 0.0.0.0:45082->45080/tcp, [::]:45082->45080/tcp

hadoo
op-hive-spark-hbase_worker2_1
35067544d3a9   nasheb/bigdata:1.0                 "/entrypoint.sh"        11 seconds ago Up 8 seconds (health:
starting)      1-4039/tcp, 0.0.0.0:4040-4043->4040-4043/tcp, [::]:4040-4043->4040-4043/tcp, 4044-8079/t
cp, 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 8081-8087/tcp, 0.0.0.0:8088->8088/tcp, [::]:8088->8
088/tcp, 8089-8999/tcp, 0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp, 9001-9867/tcp, 0.0.0.0:9868->98
68/tcp, [::]:9868->9868/tcp, 9869/tcp, 0.0.0.0:9870->9870/tcp, [::]:9870->9870/tcp, 9871-9999/tcp, 0
.0.0.0:10000->10000/tcp, [::]:10000->10000/tcp, 10001-10019/tcp, 0.0.0.0:10020->10020/tcp, [::]:1002
0->10020/tcp, 10021-16009/tcp, 0.0.0.0:16010->16010/tcp, [::]:16010->16010/tcp, 16011-18079/tcp, 0.0
.0.0:18080->18080/tcp, [::]:18080->18080/tcp, 18081-19887/tcp, 0.0.0.0:19888->19888/tcp, [::]:19888-
>19888/tcp, 19889-45079/tcp, 45081-65535/tcp, 0.0.0.0:45080->45080/tcp, [::]:45080->45080/tcp hado
```

Get HDFS ready for Nifi, with my own dataset.

```
meganpokal — meghanpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meghan...
...ata/hadoop-hive-spark-hbase — -bash
...ssh meghanpokal27@35.236.15.189
...3 meghanpokal27@35.236.15.189 ... +

hadoop-hive-spark-hbase_zoo1_1
meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$ docker-compose exec master bash
bash-5.0# start-dfs.sh
Starting namenodes on [master]
/usr/program/hadoop/bin/hdfs: /usr/program/hadoop/bin/./libexec/hadoop-config.sh: line 47: syntax error: bad substitution
Starting datanodes
/usr/program/hadoop/bin/hdfs: /usr/program/hadoop/bin/./libexec/hadoop-config.sh: line 47: syntax error: bad substitution
Starting secondary namenodes [master]
/usr/program/hadoop/bin/hdfs: /usr/program/hadoop/bin/./libexec/hadoop-config.sh: line 47: syntax error: bad substitution
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/tez/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2026-02-25 04:35:35,595 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
bash-5.0# hdfs dfs -mkdir -p /data
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/tez/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2026-02-25 04:36:55,185 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
bash-5.0# hdfs dfs -ls /
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/tez/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2026-02-25 04:37:17,260 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 7 items
drwxr-xr-x - root supergroup          0 2026-02-25 04:36 /data
drwxr-xr-x - root supergroup          0 2026-02-25 04:31 /hbase
drwxr-xr-x - root supergroup          0 2026-02-25 04:29 /log
drwxr-xr-x - root supergroup          0 2026-02-25 04:31 /spark-jars
drwxr-xr-x - root supergroup          0 2026-02-25 04:31 /tez
drwxr-xr-x - root supergroup          0 2026-02-25 04:29 /tmp
drwxrwx--- - root supergroup          0 2026-02-25 04:30 /user
bash-5.0#
```

Putting the Raw URL into the terminal.

wget <https://raw.githubusercontent.com/mvpokal/carprice650/refs/heads/main/carprices.csv> -O /tmp/carprices.csv

Using hdfs dfs -ls /data , to see if the data has been successfully imported into HDFS.


```
meganpokal — meghanpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meghan...
...ata/hadoop-hive-spark-hbase — -bash    ...ssh meghanpokal27@35.236.15.189    ...3 meghanpokal27@35.236.15.189    ... +
[meghanpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata$ ls
hadoop-hive-spark-hbase kafka nifi solr
[meghanpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata$ cd hadoop-hive-spark-hbase
[meghanpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$ docker-compose exec master bash
bash-5.0# hdfs dfs -mkdir -p /data
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/tez/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2026-02-25 04:48:41,698 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
bash-5.0# hdfs dfs -ls /
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/tez/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2026-02-25 04:48:55,861 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 7 items
drwxr-xr-x - root supergroup          0 2026-02-25 04:36 /data
drwxr-xr-x - root supergroup          0 2026-02-25 04:31 /hbase
drwxr-xr-x - root supergroup          0 2026-02-25 04:29 /log
drwxr-xr-x - root supergroup          0 2026-02-25 04:31 /spark-jars
drwxr-xr-x - root supergroup          0 2026-02-25 04:31 /tez
drwxr-xr-x - root supergroup          0 2026-02-25 04:29 /tmp
drwxrwx--- - root supergroup          0 2026-02-25 04:30 /user
```

```
meganpokal — meghanpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meghan...
...ata/hadoop-hive-spark-hbase — -bash    ...ssh meghanpokal27@35.236.15.189    ...3 meghanpokal27@35.236.15.189    ... +
drwxrwx--- - root supergroup          0 2026-02-25 04:30 /user
bash-5.0# wget https://raw.githubusercontent.com/mvpokal/carprice650/refs/heads/main/carprices.csv -O /tmp/carprices.csv
Connecting to raw.githubusercontent.com (185.199.110.133:443)
carprices.csv 100% |*****| 9741 0:00:00 ETA
bash-5.0# hdfs dfs -put /tmp/carprices.csv /data/
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/tez/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2026-02-25 04:50:28,201 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
bash-5.0# hdfs dfs -ls /data
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/tez/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2026-02-25 04:50:51,510 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
-rw-r--r-- 1 root supergroup          9741 2026-02-25 04:50 /data/carprices.csv
bash-5.0#
```

Spark Master – screenshot

spark

3.0.0

Spark Master at spark://master:7077

URL: spark://master:7077

Alive Workers: 2

Cores in use: 16 Total, 0 Used

Memory in use: 29.2 GiB Total, 0.0 B Used

Resources in use:

Applications: 0 Running, 0 Completed

Drivers: 0 Running, 0 Completed

Status: ALIVE

Workers (2)

Worker Id	Address	State	Cores	Memory	Resources
worker-20260225042925-worker1-8881	worker1-8881	ALIVE	8 (0 Used)	14.6 GiB (0.0 B Used)	
worker-20260225042925-worker2-8882	worker2-8882	ALIVE	8 (0 Used)	14.6 GiB (0.0 B Used)	

Running Applications (0)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
----------------	------	-------	---------------------	------------------------	----------------	------	-------	----------

Completed Applications (0)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
----------------	------	-------	---------------------	------------------------	----------------	------	-------	----------

Objective 2 – Hive Managed Table:

Going into the master container, hadoop-hive-spark-hbase, and starting Hive.

docker-compose exec master bash

hdfs dfs -cat /data/carprices.csv | head -n 5

Csv is loaded properly into HDFS under /data/carprices.csv

```
meganpokal — meganpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189 —...
...vue-bigdata/hadoop-hive-spark-hbase — -bash    ...k-hbase — ssh meganpokal27@35.236.15.189    ...alhost:8443 meganpokal27@35.236.15.189    ... +
meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$ docker-compose exec master bash
bash-5.0# hdfs dfs -cat /data/carprices.csv | head -n 5
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/tez/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/hive/lib/log4j-slf4j-impl-2.10.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2026-02-25 05:07:22,576 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
car_prices
year,make,model,trim,body,transmission,vin,state,condition,odometer,color,interior,seller,mmr,sellingprice,saledate
2015,Kia,Sorento,LX,SUV,automatic,5xyktca69fg566472,ca,5,16639,white,black,kia motors america inc,20500,21500,Tue
Dec 16 2014 12:30:00 GMT-0800 (PST)
2015,Kia,Sorento,LX,SUV,automatic,5xyktca69fg561319,ca,5,9393,white,beige,kia motors america inc,20800,21500,Tue D
ec 16 2014 12:30:00 GMT-0800 (PST)
2015,Volvo,S60,T5,Sedan,automatic,yv1612tb4f1310987,ca,41,14282,white,black,volvo na rep/world omni,27500,27750,Thu
Jan 29 2015 04:30:00 GMT-0800 (PST)
bash-5.0#
```

Start Hive

hive

Create Hive Table:

CREATE TABLE carprices (

```

> year INT,
> make STRING,
> model STRING,
> trim STRING,
> body STRING,
> transmission STRING,
> vin STRING,
> state STRING,
> condition INT,
> odometer INT,
> color STRING,
> interior STRING,
> seller STRING,
> mmr INT,
> sellingprice INT,
> saledate STRING
> )
> ROW FORMAT DELIMITED
> FIELDS TERMINATED BY ','
> STORED AS TEXTFILE;

```

```

meganpokal — meganpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189 — ...
...vue-bigdata/hadoop-hive-spark-hbase -- -bash  ...k-hbase -- ssh meganpokal27@35.236.15.189  ...alhost:8443 meganpokal27@35.236.15.189 ... +
hive> CREATE TABLE carprices (
>   year INT,
>   make STRING,
>   model STRING,
>   trim STRING,
>   body STRING,
>   transmission STRING,
>   vin STRING,
>   state STRING,
>   condition INT,
>   odometer INT,
>   color STRING,
>   interior STRING,
>   seller STRING,
>   mmr INT,
>   sellingprice INT,
>   saledate STRING
> )
> ROW FORMAT DELIMITED
> FIELDS TERMINATED BY ','
> STORED AS TEXTFILE;
OK
Time taken: 1.4 seconds
hive>
> █

```

Loading data from HDFS into Hive table.

```
LOAD DATA INPATH '/data/carprices.csv' INTO TABLE carprices;
```

Verifying that the data has been loaded into hive, by looking at the first 5 rows.

```
SELECT * FROM carprices LIMIT 5;
```

```
meganpokal — megalpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh megalpokal27@35.236.15.189 — ...
...vue-bigdata/hadoop-hive-spark-hbase — -bash ...k-hbase — ssh megalpokal27@35.236.15.189 ...alhost:8443 megalpokal27@35.236.15.189 +
hive>
[ > LOAD DATA INPATH '/data/carprices.csv' INTO TABLE carprices;
Loading data to table default.carprices
OK
Time taken: 0.405 seconds
hive> SELECT * FROM carprices LIMIT 5;
OK
NULL NULL NULL NULL NULL NULL NULL NULL NULL NULL NULL NULL NULL NULL
L NULL
NULL make model trim body transmission vin state NULL NULL color interior sel
ler NULL NULL saledate
2015 Kia Sorento LX SUV automatic 5xyktca69fg566472 ca 5 16639 white bla
ck kia motors america inc 20500 21500 Tue Dec 16 2014 12:30:00 GMT-0800 (PST)
2015 Kia Sorento LX SUV automatic 5xyktca69fg561319 ca 5 9393 white bei
ge kia motors america inc 20800 21500 Tue Dec 16 2014 12:30:00 GMT-0800 (PST)
2015 Volvo S60 T5 Sedan automatic yv1612tb4f1310987 ca 41 14282 white bla
ck volvo na rep/world omni 27500 27750 Thu Jan 29 2015 04:30:00 GMT-0800 (PST)
Time taken: 1.854 seconds, Fetched: 5 row(s)
```

Creating an aggregation query to confirm schema correctness.

```
SELECT make, COUNT(*) AS num_sales, AVG(sellingprice) AS avg_price
> FROM carprices
> GROUP BY make
> ORDER BY num_sales DESC;
```

```
meganpokal — megalpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh megalpokal27@35.236.15.189 — ...
...vue-bigdata/hadoop-hive-spark-hbase — -bash ...k-hbase — ssh megalpokal27@35.236.15.189 ...alhost:8443 megalpokal27@35.236.15.189 +
hive> SELECT make, COUNT(*) AS num_sales, AVG(sellingprice) AS avg_price
> FROM carprices
> GROUP BY make
[ > ORDER BY num_sales DESC;
2026-02-25 05:27:58,694 INFO [8abdf145-ec12-491e-a686-765b1a79a607 main] reducesink.VectorReduceSinkObjectHashOper
ator: VectorReduceSinkObjectHashOperator constructor vectorReduceSinkInfo org.apache.hadoop.hive.ql.plan.VectorRedu
ceSinkInfo@5d20195b
Query ID = root_20260225052758_25ec2d7d-b53c-4565-9a6f-a605d0adf6df
Total jobs = 1
Launching Job 1 out of 1
Tez session was closed. Reopening...
2026-02-25 05:27:59,196 INFO [8abdf145-ec12-491e-a686-765b1a79a607 main] client.RMProxy: Connecting to ResourceMan
ager at master/172.28.1.1:8032
Session re-established.
Session re-established.
Status: Running (Executing on YARN cluster with App id application_1771993763345_0006)

-----
VERTICES MODE STATUS TOTAL COMPLETED RUNNING PENDING FAILED KILLED
-----
Map 1 ..... container SUCCEEDED 1 1 0 0 0 0
Reducer 2 ..... container SUCCEEDED 1 1 0 0 0 0
Reducer 3 ..... container SUCCEEDED 1 1 0 0 0 0
-----
VERTICES: 03/03 [=====] 100% ELAPSED TIME: 5.24 s
-----
OK
Kia 21 22773.809523809523
Hyundai 10 8840.0
Volvo 9 30083.333333333332
Nissan 7 11085.714285714286
Chevrolet 6 33250.0
Audi 3 22266.666666666668
Ford 2 12000.0
NULL 1 NULL
Lexus 1 38400.0
make 1 NULL
Time taken: 12.038 seconds, Fetched: 10 row(s)
hive>
```


Objective 3 – Environment Setup:

Installing command packages, numpy and happybase, into the terminal.

pip3 install numpy happybase

```
meganpokal — meganpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189 —...
...vue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189      ~ — meganpokal27@dsc650bigdata: ~ — -bash      ... +
meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$ docker-compose exec master bash
bash-5.0# pip3 install numpy happybase
Collecting numpy
  Downloading numpy-1.21.6.zip (10.3 MB)
    |████████████████████████████████████████| 10.3 MB 12.3 MB/s
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing wheel metadata ... done
Collecting happybase
  Downloading happybase-1.3.0-py2.py3-none-any.whl (26 kB)
Collecting importlib-resources
  Downloading importlib_resources-5.12.0-py3-none-any.whl (36 kB)
Collecting six
  Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Collecting thriftypy2>=0.4
  Downloading thriftypy2-0.6.0.tar.gz (996 kB)
    |████████████████████████████████████████| 996 kB 31.3 MB/s
  Installing build dependencies ... done
  WARNING: Missing build requirements in pyproject.toml for thriftypy2>=0.4 from https://files.pythonhosted.org/packages/cc/7c/bc038fb868b6b5474101197d42866fbc61839c4f739816587a748ac68541/thriftypy2-0.6.0.tar.gz#sha256=621c263f99274d51a9a1deecf301845f1408d497bdafed682db6155132a99cf4 (from happybase).
  WARNING: The project does not specify a build backend, and pip cannot fall back to setuptools without 'wheel'.
  Getting requirements to build wheel ... done
  Installing backend dependencies ... done
  Preparing wheel metadata ... done
Collecting zipp>=3.1.0; python_version < "3.10"
  Downloading zipp-3.15.0-py3-none-any.whl (6.8 kB)
Collecting typing-extensions>=3.7.4; python_version < "3.8"
  Downloading typing_extensions-4.7.1-py3-none-any.whl (33 kB)
Collecting ply<4.0,>=3.4
  Downloading ply-3.11-py2.py3-none-any.whl (49 kB)
    |████████████████████████████████████████| 49 kB 8.3 MB/s
Building wheels for collected packages: numpy, thriftypy2
  Building wheel for numpy (PEP 517) ... /
done
  Created wheel for numpy: filename=numpy-1.21.6-cp37-cp37m-linux_x86_64.whl size=16643972 sha256=850ce5edb44b4c218c35278b28e2b3497c23711ce3858acd2dd11704cee32799
  Stored in directory: /root/.cache/pip/wheels/4e/7e/9e/0fde042ccff2493994076dac9c3fbd78feb444c3bd94eb386a
  Building wheel for thriftypy2 (PEP 517) ... |
```



```
meganpokal — meganpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189 — 1...
...evue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189      ~ — meganpokal27@dsc650bigdata: ~ — -bash      ... +
meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$ docker-compose exec worker1 bash
bash-5.0# pip3 install numpy happybase
Collecting numpy
  Downloading numpy-1.21.6.zip (10.3 MB)
    |████████████████████████████████████████| 10.3 MB 12.4 MB/s
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing wheel metadata ... done
Collecting happybase
  Downloading happybase-1.3.0-py2.py3-none-any.whl (26 kB)
Collecting importlib-resources
  Downloading importlib_resources-5.12.0-py3-none-any.whl (36 kB)
Collecting six
  Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Collecting thriftypy2>=0.4
  Downloading thriftypy2-0.6.0.tar.gz (996 kB)
    |████████████████████████████████████████| 996 kB 31.8 MB/s
  Installing build dependencies ... done
  WARNING: Missing build requirements in pyproject.toml for thriftypy2>=0.4 from https://files.pythonhosted.org/packages/cc/7c/bc038fb868b6b5474101197d42866fbc61839c4f739816587a748ac68541/thriftypy2-0.6.0.tar.gz#sha256=621c263f99274d51a9a1deecf301845f1408d497bdafed682db6155132a99cf4 (from happybase).
  WARNING: The project does not specify a build backend, and pip cannot fall back to setuptools without 'wheel'.
  Getting requirements to build wheel ... done
  Installing backend dependencies ... done
  Preparing wheel metadata ... done
Collecting zipp>=3.1.0; python_version < "3.10"
  Downloading zipp-3.15.0-py3-none-any.whl (6.8 kB)
Collecting typing-extensions>=3.7.4; python_version < "3.8"
  Downloading typing_extensions-4.7.1-py3-none-any.whl (33 kB)
Collecting ply<4.0,>=3.4
  Downloading ply-3.11-py2.py3-none-any.whl (49 kB)
    |████████████████████████████████████████| 49 kB 8.1 MB/s
Building wheels for collected packages: numpy, thriftypy2
  Building wheel for numpy (PEP 517) ... |
-
done
  Created wheel for numpy: filename=numpy-1.21.6-cp37-cp37m-linux_x86_64.whl size=16643977 sha256=c6bc4086654886bba4390173b7f12b40a0a54dc9bd29265af4ccf93b33634700
```

```
meganpokal — meganpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189 — 1...
...evue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189 ~ — meganpokal27@dsc650bigdata: ~ — -bash ... +

Installing build dependencies ... done
WARNING: Missing build requirements in pyproject.toml for thriftypy2>=0.4 from https://files.pythonhosted.org/packages/cc/7c/bc038fb868b6b5474101197d42866fbc61839c4f739816587a748ac68541/thriftypy2-0.6.0.tar.gz#sha256=621c263f99274d51a9a1deecf301845f1408d497bdafed682db6155132a99cf4 (from happybase).
WARNING: The project does not specify a build backend, and pip cannot fall back to setuptools without 'wheel'.
Getting requirements to build wheel ... done
Installing backend dependencies ... done
Preparing wheel metadata ... done
Collecting zipp>=3.1.0; python_version < "3.10"
  Downloading zipp-3.15.0-py3-none-any.whl (6.8 kB)
Collecting typing-extensions>=3.7.4; python_version < "3.8"
  Downloading typing_extensions-4.7.1-py3-none-any.whl (33 kB)
Collecting ply<4.0,>=3.4
  Downloading ply-3.11-py2.py3-none-any.whl (49 kB)
Building wheels for collected packages: numpy, thriftypy2
  Building wheel for numpy (PEP 517) ... |
-
done
  Created wheel for numpy: filename=numpy-1.21.6-cp37-cp37m-linux_x86_64.whl size=16643977 sha256=c6bc4086654886bba4390173b7f12b40a0a54dc9bd29265af4ccf93b33634700
  Stored in directory: /root/.cache/pip/wheels/4e/7e/9e/0fde042ccff2493994076dac9c3fbd78feb444c3bd94eb386a
  Building wheel for thriftypy2 (PEP 517) ... \
done
  Created wheel for thriftypy2: filename=thriftypy2-0.6.0-cp37-cp37m-linux_x86_64.whl size=1479788 sha256=7378e6c419fe83fced7dcdd197e15ad9bd41cab7873dc368bce75ee49fba945e
  Stored in directory: /root/.cache/pip/wheels/ce/e1/d3/38770f98821c966d3ad3bb392293b6e0bc189e89580338db63
Successfully built numpy thriftypy2
Installing collected packages: numpy, zipp, importlib-resources, six, typing-extensions, ply, thriftypy2, happybase
Successfully installed happybase-1.3.0 importlib-resources-5.12.0 numpy-1.21.6 ply-3.11 six-1.17.0 thriftypy2-0.6.0 t
yping-extensions-4.7.1 zipp-3.15.0
bash-5.0#
bash-5.0#
bash-5.0#
bash-5.0#
bash-5.0# exit
exit
meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$
```

Worker 2:

```
Docker-compose exec worker2 bash
Pip3 install numpy happybase
exit
```



```

meganpokal — meganpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189 — 1...
...ark-hbase — ssh meganpokal27@35.236.15.189  ~ — meganpokal27@dsc650bigdata: ~ — -bash  ...alhost:8443 meganpokal27@35.236.15.189  +
meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$ docker-compose exec worker2 bash
bash-5.0# pip3 install numpy happybase
Collecting numpy
  Downloading numpy-1.21.6.zip (10.3 MB)
    |████████████████████████████████████████| 10.3 MB 11.4 MB/s
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing wheel metadata ... done
Collecting happybase
  Downloading happybase-1.3.0-py2.py3-none-any.whl (26 kB)
Collecting importlib-resources
  Downloading importlib_resources-5.12.0-py3-none-any.whl (36 kB)
Collecting six
  Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Collecting thriftypy2>=0.4
  Downloading thriftypy2-0.6.0.tar.gz (996 kB)
    |████████████████████████████████████████| 996 kB 24.1 MB/s
  Installing build dependencies ... done
  WARNING: Missing build requirements in pyproject.toml for thriftypy2>=0.4 from https://files.pythonhosted.org/packages/cc/7c/bc038fb868b6b5474101197d42866fbc61839c4f739816587a748ac68541/thriftypy2-0.6.0.tar.gz#sha256=621c263f99274d51a9a1deecf301845f1408d497bdafed682db6155132a99cf4 (from happybase).
  WARNING: The project does not specify a build backend, and pip cannot fall back to setuptools without 'wheel'.
  Getting requirements to build wheel ... done
  Installing backend dependencies ... done
  Preparing wheel metadata ... done
Collecting zipp>=3.1.0; python_version < "3.10"
  Downloading zipp-3.15.0-py3-none-any.whl (6.8 kB)
Collecting typing-extensions>=3.7.4; python_version < "3.8"
  Downloading typing_extensions-4.7.1-py3-none-any.whl (33 kB)
Collecting ply<4.0,>=3.4
  Downloading ply-3.11-py2.py3-none-any.whl (49 kB)
    |████████████████████████████████████████| 49 kB 7.6 MB/s
Building wheels for collected packages: numpy, thriftypy2
  Building wheel for numpy (PEP 517) ... /
done
  Created wheel for numpy: filename=numpy-1.21.6-cp37-cp37m-linux_x86_64.whl size=16644652 sha256=69dc301764c3bc58e6331ff988874ea462672b6b5f2f7c8e6f3df00ecd4aba8c
  Stored in directory: /root/.cache/pip/wheels/4e/7e/9e/0fde042ccff2493994076dac9c3fbd78feb444c3bd94eb386a
  Building wheel for thriftypy2 (PEP 517) ... done

```

```
meganpokal — meganpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189 — 1...
...ark-hbase — ssh meganpokal27@35.236.15.189  ~ — meganpokal27@dsc650bigdata: ~ — -bash  ...alhost:8443 meganpokal27@35.236.15.189  +
Downloading importlib_resources-5.12.0-py3-none-any.whl (36 kB)
Collecting six
  Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Collecting thriftypy2>=0.4
  Downloading thriftypy2-0.6.0.tar.gz (996 kB)
    |████████████████████████████████████████| 996 kB 24.1 MB/s
Installing build dependencies ... done
WARNING: Missing build requirements in pyproject.toml for thriftypy2>=0.4 from https://files.pythonhosted.org/packages/cc/7c/bc038fb868b6b5474101197d42866fbc61839c4f739816587a748ac68541/thriftypy2-0.6.0.tar.gz#sha256=621c263f99274d51a9a1deecf301845f1408d497bdafed682db6155132a99cf4 (from happybase).
WARNING: The project does not specify a build backend, and pip cannot fall back to setuptools without 'wheel'.
Getting requirements to build wheel ... done
Installing backend dependencies ... done
Preparing wheel metadata ... done
Collecting zipp>=3.1.0; python_version < "3.10"
  Downloading zipp-3.15.0-py3-none-any.whl (6.8 kB)
Collecting typing_extensions>=3.7.4; python_version < "3.8"
  Downloading typing_extensions-4.7.1-py3-none-any.whl (33 kB)
Collecting ply<4.0,>=3.4
  Downloading ply-3.11-py2.py3-none-any.whl (49 kB)
    |████████████████████████████████████████| 49 kB 7.6 MB/s
Building wheels for collected packages: numpy, thriftypy2
Building wheel for numpy (PEP 517) ... /
done
  Created wheel for numpy: filename=numpy-1.21.6-cp37-cp37m-linux_x86_64.whl size=16644652 sha256=69dc301764c3bc58e6331ff988874ea462672b6b5f2f7c8e6f3df00ecd4aba8c
  Stored in directory: /root/.cache/pip/wheels/4e/7e/9e/0fde042ccff2493994076dac9c3fbd78feb444c3bd94eb386a
  Building wheel for thriftypy2 (PEP 517) ... done
  Created wheel for thriftypy2: filename=thriftypy2-0.6.0-cp37-cp37m-linux_x86_64.whl size=1479769 sha256=728051b4e95f82a31127dd71515a86b0c99678a0f6dc16734e9fc38f86f2647f
  Stored in directory: /root/.cache/pip/wheels/ce/e1/d3/38770f98821c966d3ad3bb392293b6e0bc189e89580338db63
Successfully built numpy thriftypy2
Installing collected packages: numpy, zipp, importlib-resources, six, typing_extensions, ply, thriftypy2, happybase
Successfully installed happybase-1.3.0 importlib-resources-5.12.0 numpy-1.21.6 ply-3.11 six-1.17.0 thriftypy2-0.6.0 t
yping_extensions-4.7.1 zipp-3.15.0
bash-5.0#
bash-5.0# exit
exit
meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$
```

Starting the HBase Thrift Server:

I will open the master container;
docker-compose exec master bash
Start-hbase.sh
hbase thrift srtart -threadpool

```

meganpokal — meganpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189 — 1...
...ark-hbase — ssh meganpokal27@35.236.15.189  ~ -- meganpokal27@dsc650bigdata: ~ -- -bash  ...alhost:8443 meganpokal27@35.236.15.189  +
exit
meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$ docker-compose exec master bash
bash-5.0# start-hbase.sh
master running as process 1229. Stop it first.
worker2: running regionserver, logging to /usr/program/hbase/bin/./logs/hbase-root-regionserver-worker2.out
worker1: running regionserver, logging to /usr/program/hbase/bin/./logs/hbase-root-regionserver-worker1.out
bash-5.0# hbase thrift start -threadpool
2026-03-02 23:32:23,809 INFO [main] thrift.ThriftServer: ***** STARTING service 'ThriftServer' *****
2026-03-02 23:32:23,812 INFO [main] util.VersionInfo: HBase 2.3.6
2026-03-02 23:32:23,812 INFO [main] util.VersionInfo: Source code repository git://7257be94c61e/home/stack/hbase-rm
/output/hbase revision=7414579f2620fca6b75146c29ab2726fc4643ac9
2026-03-02 23:32:23,812 INFO [main] util.VersionInfo: Compiled by stack on Wed Jul 28 22:24:42 UTC 2021
2026-03-02 23:32:23,812 INFO [main] util.VersionInfo: From source with checksum eb11554aeb325ccaf65a9b469efe7ddcdc0
bb0eec35a53f77d0c5c09eac5722007a05ef62349c7afb8005ed1abe32e772f99bb2137194641f5ce51d525aa5634
2026-03-02 23:32:24,074 INFO [main] thrift.ImplType: Using thrift server type threadpool
2026-03-02 23:32:24,106 WARN [main] util.NativeCodeLoader: Unable to load native-hadoop library for your platform..
. using builtin-java classes where applicable
2026-03-02 23:32:24,504 INFO [main] metrics.MetricRegistries: Loaded MetricRegistries class org.apache.hadoop.hbase
.metrics.impl.MetricRegistriesImpl
2026-03-02 23:32:24,553 INFO [main] hbase.ChoreService: Chore ScheduledChore name=ConnectionCleaner, period=10000,
unit=MILLISECONDS is enabled.
2026-03-02 23:32:24,624 INFO [main] thrift.ThriftServer: starting HBase ThreadPool Thrift server on /0.0.0.0:9090
2026-03-02 23:32:24,650 INFO [main] util.log: Logging initialized @1171ms
2026-03-02 23:32:24,738 INFO [main] http.HttpServer: Added global filter 'safety' (class=org.apache.hadoop.hbase.ht
tp.HttpServer$QuotingInputFilter)
2026-03-02 23:32:24,738 INFO [main] http.HttpServer: Added global filter 'clickjackingprevention' (class=org.apache
.hadoop.hbase.http.ClickjackingPreventionFilter)
2026-03-02 23:32:24,739 INFO [main] http.HttpServer: Added global filter 'securityheaders' (class=org.apache.hadoop
.hbase.http.SecurityHeadersFilter)
2026-03-02 23:32:24,742 INFO [main] http.HttpServer: Added filter static_user_filter (class=org.apache.hadoop.hbase
.http.lib.StaticUserWebFilter$StaticUserFilter) to context thrift
2026-03-02 23:32:24,742 INFO [main] http.HttpServer: Added filter static_user_filter (class=org.apache.hadoop.hbase
.http.lib.StaticUserWebFilter$StaticUserFilter) to context logs
2026-03-02 23:32:24,742 INFO [main] http.HttpServer: Added filter static_user_filter (class=org.apache.hadoop.hbase
.http.lib.StaticUserWebFilter$StaticUserFilter) to context static
2026-03-02 23:32:24,750 INFO [main] http.HttpServer: ASYNC_PROFILER_HOME environment variable and async.profiler.ho
me system property not specified. Disabling /prof endpoint.
2026-03-02 23:32:24,789 INFO [main] http.HttpServer: Jetty bound to port 9095
2026-03-02 23:32:24,790 INFO [main] server.Server: jetty-9.3.28.v20191105, build timestamp: 2019-11-05T18:00:51Z, g

```



```
meganpokal — meganpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189 — 1...
...ark-hbase — ssh meganpokal27@35.236.15.189  ~ -- meganpokal27@dsc650bigdata: ~ -- -bash  ...alhost:8443 meganpokal27@35.236.15.189  +
2026-03-02 23:32:24,553 INFO [main] hbase.ChoreService: Chore ScheduledChore name=ConnectionCleaner, period=10000,
unit=MILLISECONDS is enabled.
2026-03-02 23:32:24,624 INFO [main] thrift.ThriftServer: starting HBase ThreadPool Thrift server on /0.0.0.0:9090
2026-03-02 23:32:24,650 INFO [main] util.log: Logging initialized @1171ms
2026-03-02 23:32:24,738 INFO [main] http.HttpServer: Added global filter 'safety' (class=org.apache.hadoop.hbase.ht
tp.HttpServer$QuotingInputFilter)
2026-03-02 23:32:24,738 INFO [main] http.HttpServer: Added global filter 'clickjackingprevention' (class=org.apache
.hadoop.hbase.http.ClickjackingPreventionFilter)
2026-03-02 23:32:24,739 INFO [main] http.HttpServer: Added global filter 'securityheaders' (class=org.apache.hadoop
.hbase.http.SecurityHeadersFilter)
2026-03-02 23:32:24,742 INFO [main] http.HttpServer: Added filter static_user_filter (class=org.apache.hadoop.hbase
.http.lib.StaticUserWebFilter$StaticUserFilter) to context thrift
2026-03-02 23:32:24,742 INFO [main] http.HttpServer: Added filter static_user_filter (class=org.apache.hadoop.hbase
.http.lib.StaticUserWebFilter$StaticUserFilter) to context logs
2026-03-02 23:32:24,742 INFO [main] http.HttpServer: Added filter static_user_filter (class=org.apache.hadoop.hbase
.http.lib.StaticUserWebFilter$StaticUserFilter) to context static
2026-03-02 23:32:24,750 INFO [main] http.HttpServer: ASYNC_PROFILER_HOME environment variable and async.profiler.ho
me system property not specified. Disabling /prof endpoint.
2026-03-02 23:32:24,789 INFO [main] http.HttpServer: Jetty bound to port 9095
2026-03-02 23:32:24,790 INFO [main] server.Server: jetty-9.3.28.v20191105, build timestamp: 2019-11-05T18:00:51Z, g
it hash: d7dd68d6e9d8ff06a0130e46886c2db5d70784c1
2026-03-02 23:32:24,823 INFO [main] http.SecurityHeadersFilter: Added security headers filter
2026-03-02 23:32:24,827 INFO [main] handler.ContextHandler: Started o.e.j.s.ServletContextHandler@6e9a5ed8{/logs,fi
le:///usr/program/hbase/logs/,AVAILABLE}
2026-03-02 23:32:24,827 INFO [main] http.SecurityHeadersFilter: Added security headers filter
2026-03-02 23:32:24,827 INFO [main] handler.ContextHandler: Started o.e.j.s.ServletContextHandler@6c284af{/static,f
ile:///usr/program/hbase/hbase-webapps/static/,AVAILABLE}
2026-03-02 23:32:24,933 INFO [main] webapp.StandardDescriptorProcessor: NO JSP Support for /, did not find org.ecli
pse.jetty.jsp.JettyJspServlet
2026-03-02 23:32:24,942 INFO [main] http.SecurityHeadersFilter: Added security headers filter
2026-03-02 23:32:24,954 INFO [main] handler.ContextHandler: Started o.e.j.w.WebAppContext@205d38da{/,file:///usr/pr
ogram/hbase/hbase-webapps/thrift/,AVAILABLE}{file:///usr/program/hbase/hbase-webapps/thrift}
2026-03-02 23:32:24,961 INFO [main] server.AbstractConnector: Started ServerConnector@51a9ad5e{HTTP/1.1,[http/1.1]}
{0.0.0.0:9095}
2026-03-02 23:32:24,961 INFO [main] server.Server: Started @1482ms
```

The installation of numpy and happybase is required to enable Python based data processing and communication with Hbase. NumPy provides excellent number evaluation that is commonly used in data analysis and machine learning workflows. HappyBase allows Python applications to interact with HBase through the Thrift server, starting the Spark Machine Learning job that will evaluate metrics such as RMSE and R2 into the HBase table.

Objective 4 – HBase Table Creation:

Creating the command for the HBase table.

While Thrift is running in one terminal, I will open a new tab to run Objective 4.

Hbase shell


```
meganpokal — meganpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189 — 1...
...h meganpokal27@35.236.15.189 ...h meganpokal27@35.236.15.189 ...7@dsc650bigdata: ~ — -bash ...meganpokal27@35.236.15.189 +
Last login: Mon Mar  2 23:27:06 2026 from 174.233.226.84
meganpokal27@dsc650bigdata:~$ ls
dsc650-infra
meganpokal27@dsc650bigdata:~$ cd dsc650-infra/bellevue-bigdata
meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata$ ls
hadoop-hive-spark-hbase kafka hif solr
meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata$ cd hadoop-hive-spark-hbase
meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$ docker-compose exec master bash
bash-5.0# hbase shell
2026-03-02 23:54:19,843 WARN [main] util.NativeCodeLoader: Unable to load native-hadoop library for your platform..
. using builtin-java classes where applicable
HBase Shell
Use "help" to get list of supported commands.
Use "exit" to quit this interactive shell.
For Reference, please visit: http://hbase.apache.org/2.0/book.html#shell
Version 2.3.6, r7414579f2620fca6b75146c29ab2726fc4643ac9, Wed Jul 28 22:24:42 UTC 2021
Took 0.0013 seconds
hbase(main):001:0>
```

Creating the table from my Car dataset.

Table:

```
create 'car_sales', 'details'
```

The table `car_sales` has been created with a single column family named `details`. Using one column family improves storage efficiency and keeps related groups together, improving the read performance.

```
meganpokal — meganpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189 — 1...
...h meganpokal27@35.236.15.189 ...h meganpokal27@35.236.15.189 ...7@dsc650bigdata: ~ — -bash ...meganpokal27@35.236.15.189 +
Version 2.3.6, r7414579f2620fca6b75146c29ab2726fc4643ac9, Wed Jul 28 22:24:42 UTC 2021
Took 0.0013 seconds
hbase(main):001:0> create 'car_sales', 'details'
Created table car_sales
Took 1.7001 seconds
=> Hbase::Table - car_sales
hbase(main):002:0>
```

Empty Scan from the table that was created, `car_sales`

Meaning that the table has been created but there is no data in the table yet.

```
meganpokal — meganpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189 — 1...
...h meganpokal27@35.236.15.189 ...h meganpokal27@35.236.15.189 ...7@dsc650bigdata: ~ — -bash ...meganpokal27@35.236.15.189 +
Took 1.7001 seconds
=> Hbase::Table - car_sales
hbase(main):002:0> scan 'car_sales'
ROW COLUMN+CELL
0 row(s)
Took 0.1649 seconds
hbase(main):003:0>
```

Row Key Design:

The row key is designed to identify each vehicle sale record. Making each row unique, and efficient lookup since HBase indexes and sorts by row key. Column name will be the details.

'car_sales' → table name

'wbsfv9c51ed593089' → row key; VIN of vehicle

'details:year' → column family + column; year of the car

'2014' → stored value; year of the car from the dataset

All of these under the same row key, will belong to the same vehicle.

```
put 'car_sales', 'wbsfv9c51ed593089', 'details:year', '2014'
put 'car_sales', 'wbsfv9c51ed593089', 'details:make', 'BMW'
put 'car_sales', 'wbsfv9c51ed593089', 'details:model', 'M5'
put 'car_sales', 'wbsfv9c51ed593089', 'details:state', 'ca'
```

Then scan to make sure that it has been applied successfully. 1 row has been made!

Scan 'car_sales'



```
meganpokal — meganpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189 — 1...
...h meganpokal27@35.236.15.189      ...h meganpokal27@35.236.15.189      ...7@dsc650bigdata: ~ — -bash      ...meganpokal27@35.236.15.189      +

hbase(main):004:0> put 'car_sales', 'wbsfv9c51ed593089', 'details:year', '2014'
Took 0.0487 seconds
hbase(main):005:0> put 'car_sales', 'wbsfv9c51ed593089', 'details:make', 'BMW'
Took 0.0064 seconds
hbase(main):006:0> put 'car_sales', 'wbsfv9c51ed593089', 'details:model', 'M5'
Took 0.0075 seconds
hbase(main):007:0> put 'car_sales', 'wbsfv9c51ed593089', 'details:state', 'ca'
Took 0.0077 seconds
hbase(main):008:0> scan 'car_sales'
ROW                                COLUMN+CELL
wbsfv9c51ed593089                  column=details:make, timestamp=2026-03-03T00:09:13.325, value=BMW
wbsfv9c51ed593089                  column=details:model, timestamp=2026-03-03T00:09:13.355, value=M5
wbsfv9c51ed593089                  column=details:state, timestamp=2026-03-03T00:09:16.662, value=ca
wbsfv9c51ed593089                  column=details:year, timestamp=2026-03-03T00:09:13.286, value=2014
1 row(s)
Took 0.0387 seconds
hbase(main):009:0>
```

Objective 5 – PySpark ML Code:

Go to the master container and run PySpark.

Having the Thrift server running on the other terminal.

Open Visual Studio Code, to make a new file named car_prices_ml.py

Code in VS code:

File name: car_prices_ml.py

```
from pyspark.sql import SparkSession
```

```

from pyspark.ml.feature import VectorAssembler
from pyspark.ml.regression import LinearRegression
import happybase

# Spark session
spark = SparkSession.builder.appName("CarPricesML").enableHiveSupport().getOrCreate()

# Load data from Hive
df = spark.sql("""
    SELECT year, odometer, mmr, sellingprice
    FROM carprices
""")

# Handle nulls
df = df.na.drop()

# Assemble features
assembler = VectorAssembler(inputCols=["year", "odometer", "mmr"], outputCol="features",
                             handleInvalid="skip")
assembled_df = assembler.transform(df).select("features", "sellingprice")

# Split into train/test
train_data, test_data = assembled_df.randomSplit([0.7, 0.3])

# Train Linear Regression
lr = LinearRegression(labelCol="sellingprice")
lr_model = lr.fit(train_data)

# Evaluate
results = lr_model.evaluate(test_data)
print(f'RMSE: {results.rootMeanSquaredError}')
print(f'R2: {results.r2}')

# Write metrics to HBase
data = [
    ('metrics1', 'details:rmse', str(results.rootMeanSquaredError)),
    ('metrics1', 'details:r2', str(results.r2))
]

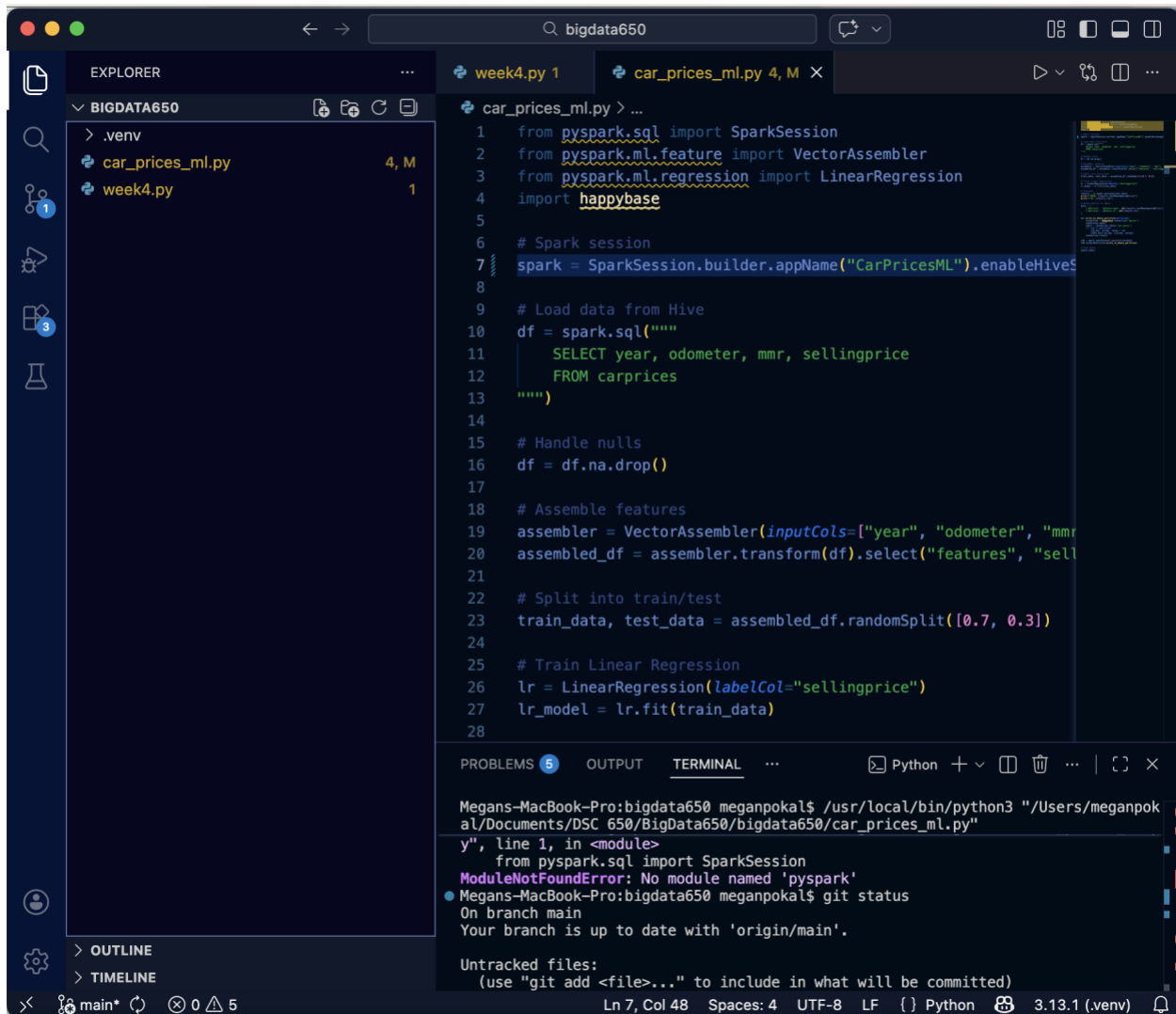
def write_to_hbase_partition(partition):
    connection = happybase.Connection('master')
    connection.open()
    table = connection.table('car_sales')
    for row in partition:
        row_key, column, value = row
        table.put(row_key, {column: value})

```

```
connection.close()
```

```
rdd = spark.sparkContext.parallelize(data)  
rdd.foreachPartition(write_to_hbase_partition)
```

```
# Stop Spark  
spark.stop()
```



The screenshot shows the Visual Studio Code interface. The Explorer panel on the left shows a project named 'BIGDATA650' with files '.venv', 'car_prices_ml.py' (4, M), and 'week4.py' (1). The main editor window displays the 'car_prices_ml.py' file with the following code:

```
1 from pyspark.sql import SparkSession  
2 from pyspark.ml.feature import VectorAssembler  
3 from pyspark.ml.regression import LinearRegression  
4 import happybase  
5  
6 # Spark session  
7 spark = SparkSession.builder.appName("CarPricesML").enableHiveSupport().getOrCreate()  
8  
9 # Load data from Hive  
10 df = spark.sql("""  
11     SELECT year, odometer, mmr, sellingprice  
12     FROM carprices  
13 """)  
14  
15 # Handle nulls  
16 df = df.na.drop()  
17  
18 # Assemble features  
19 assembler = VectorAssembler(inputCols=["year", "odometer", "mmr", "sellingprice"])  
20 assembled_df = assembler.transform(df).select("features", "sellingprice")  
21  
22 # Split into train/test  
23 train_data, test_data = assembled_df.randomSplit([0.7, 0.3])  
24  
25 # Train Linear Regression  
26 lr = LinearRegression(labelCol="sellingprice")  
27 lr_model = lr.fit(train_data)  
28
```

The terminal window at the bottom shows the following output:

```
Megans-MacBook-Pro:bigdata650 meganpokal$ /usr/local/bin/python3 "/Users/meganpokal/Documents/DSC 650/BigData650/bigdata650/car_prices_ml.py"  
y", line 1, in <module>  
    from pyspark.sql import SparkSession  
ModuleNotFoundError: No module named 'pyspark'  
● Megans-MacBook-Pro:bigdata650 meganpokal$ git status  
On branch main  
Your branch is up to date with 'origin/main'.  
  
Untracked files:  
  (use "git add <file>..." to include in what will be committed)  
Ln 7, Col 48 Spaces: 4 UTF-8 LF {} Python 3.13.1 (.venv)
```

Inside of VS Code, terminal, I was trying to add this file to the GitHub repository. I had to install pyspark first. Then add the file to GitHub through VS Code.

```
Git add car_prices_ml.py
```

```
Ls – successfully shows both files in the repository
```


mvpokal / bigdata650

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bigdata650

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main

Go to file

Code

mvpokal

Add PySpark ML script for car prices

b6082e3 · 3 minutes ago

car_prices_ml.py

Add PySpark ML script for car prices

3 minutes ago

week4.py

save to me

2 months ago

README

Add a README

Help people interested in this repository understand your project.

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No description, website, or topics provided.

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Create a new release

Packages

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Contributors 1

mvpokal

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main

bigdata650 / car_prices_ml.py

Go to file

mvpokal

Fix app name in Spark session initialization

18b36e8 · now

53 lines (42 loc) · 1.44 KB

Code

Blame

Raw

Download

Copy

Share

```
1 from pyspark.sql import SparkSession
2 from pyspark.ml.feature import VectorAssembler
3 from pyspark.ml.regression import LinearRegression
4 import happybase
5
6 # Spark session
7 spark = SparkSession.builder.appName("CarPricesML").enableHiveSupport().getOrCreate()
8
9 # Load data from Hive
10 df = spark.sql("""
11     SELECT year, odometer, mmr, sellingprice
12     FROM carprices
13 """)
14
15 # Handle nulls
16 df = df.na.drop()
17
18 # Assemble features
19 assembler = VectorAssembler(inputCols=["year", "odometer", "mmr"], outputCol="features", handleInvalid="skip")
20 assembled_df = assembler.transform(df).select("features", "sellingprice")
21
22 # Split into train/test
23 train_data, test_data = assembled_df.randomSplit([0.7, 0.3])
```

In the terminal, in the master container, I will enter the Raw GitHub link to put the new file `car_prices_ml.py`

```
meganpokal — megalpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh megalpokal27@35.236.15.189 — 1...
...pokal27@35.236.15.189 ...pokal27@35.236.15.189 ...oop-hive-spark-hbase ...spark-hbase — -bash ...kal27@35.236.15.189 +
meganpokal27@dsc650bigdata:~$ cd dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase
meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$ ls
check_child_image.sh data_share docker-compose-up-logging.sh init.sql logs new_york_tree_species.csv rm_none_images.sh
data docker-compose-logging.sh init.sql logs nohup.out tail_docker_compose_logs.sh
meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$ docker-compose exec master bash
lbash-5.0# cd /usr/program/source_dir
lbash-5.0# curl -O https://raw.githubusercontent.com/mvpokal/bigdata650/refs/heads/main/car_prices_ml.py
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 1476 100 1476 0 0 10179 0 --:--:-- --:--:-- --:--:-- 10179
lbash-5.0# ls
car_price_ml.py car_prices_ml.py
lbash-5.0#
```

Run PySpark:

`spark-submit \`

`> --master yarn \`

`> --deploy-mode client \`

`> --name CarPricesML \`

`> car_prices_ml.py`

```
meganpokal — megalpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh megalpokal27@35.236.15.189 — 1...
...pokal27@35.236.15.189 ...pokal27@35.236.15.189 ...oop-hive-spark-hbase ...spark-hbase — -bash ...kal27@35.236.15.189 +
bash-5.0# spark-submit \
> --master yarn \
> --deploy-mode client \
> --name CarPricesML \
> car_prices_ml.py
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/spark/jars/slf4j-log4j12-1.7.30.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
0 [main] WARN org.apache.hadoop.util.NativeCodeLoader - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
1129 [Thread-5] INFO org.apache.spark.SparkContext - Running Spark version 3.0.0
1179 [Thread-5] INFO org.apache.spark.resource.ResourceUtils - =====
1180 [Thread-5] INFO org.apache.spark.resource.ResourceUtils - Resources for spark.driver:

1180 [Thread-5] INFO org.apache.spark.resource.ResourceUtils - =====
1181 [Thread-5] INFO org.apache.spark.SparkContext - Submitted application: CarPriceML
1279 [Thread-5] INFO org.apache.spark.SecurityManager - Changing view acls to: root
1279 [Thread-5] INFO org.apache.spark.SecurityManager - Changing modify acls to: root
1279 [Thread-5] INFO org.apache.spark.SecurityManager - Changing view acls groups to:
1279 [Thread-5] INFO org.apache.spark.SecurityManager - Changing modify acls groups to:
1279 [Thread-5] INFO org.apache.spark.SecurityManager - SecurityManager: authentication disabled; ui acls disabled; users with view permissions: S
et(root); groups with view permissions: Set(); users with modify permissions: Set(root); groups with modify permissions: Set()
1587 [Thread-5] INFO org.apache.spark.util.Utils - Successfully started service 'sparkDriver' on port 7001.
1634 [Thread-5] INFO org.apache.spark.SparkEnv - Registering MapOutputTracker
1686 [Thread-5] INFO org.apache.spark.SparkEnv - Registering BlockManagerMaster
1709 [Thread-5] INFO org.apache.spark.storage.BlockManagerMasterEndpoint - Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
1710 [Thread-5] INFO org.apache.spark.storage.BlockManagerMasterEndpoint - BlockManagerMasterEndpoint up
1760 [Thread-5] INFO org.apache.spark.SparkEnv - Registering BlockManagerMasterHeartbeat
1785 [Thread-5] INFO org.apache.spark.storage.DiskBlockManager - Created local directory at /tmp/blockmgr-30561788-f6f0-45a5-b8e7-1d3664eecccd
1818 [Thread-5] INFO org.apache.spark.storage.memory.MemoryStore - MemoryStore started with capacity 366.3 MiB
1871 [Thread-5] INFO org.apache.spark.SparkEnv - Registering OutputCommitCoordinator
2005 [Thread-5] INFO org.sparkproject.jetty.util.log - Logging initialized @3668ms to org.sparkproject.jetty.util.log.Slf4jLog
2108 [Thread-5] INFO org.sparkproject.jetty.server.Server - jetty-9.4.z-SNAPSHOT; built: 2019-04-29T20:42:08.989Z; git: e1bc35120a6617ee3df052294e433f3a25ce7097; jvm 1.8.0_275-b01
```

Showing the `car_sales` table and the ML code.

Hbase shell

Scan 'car_sales'

PySpark ML has run successfully and wrote the evaluation to HBase.

```
meganpokal — megalpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh megalpokal27@35.236.15.189 — 1...
...pokal27@35.236.15.189 ... ..pokal27@35.236.15.189 ...oop-hive-spark-hbase ...spark-hbase — -bash ...kal27@35.236.15.189 +
55545 [YARN application state monitor] INFO org.apache.spark.scheduler.cluster.YarnClientSchedulerBackend - Interrupting monitor thread
55569 [Thread-5] INFO org.apache.spark.scheduler.cluster.YarnClientSchedulerBackend - Shutting down all executors
55571 [dispatcher-CoarseGrainedScheduler] INFO org.apache.spark.scheduler.cluster.YarnSchedulerBackend$YarnDriverEndpoint - Asking each executor to
shut down
55578 [Thread-5] INFO org.apache.spark.scheduler.cluster.YarnClientSchedulerBackend - YARN client scheduler backend Stopped
56027 [dispatcher-event-loop-0] INFO org.apache.spark.MapOutputTrackerMasterEndpoint - MapOutputTrackerMasterEndpoint stopped!
56045 [Thread-5] INFO org.apache.spark.storage.memory.MemoryStore - MemoryStore cleared
56046 [Thread-5] INFO org.apache.spark.storage.BlockManager - BlockManager stopped
56048 [Thread-5] INFO org.apache.spark.storage.BlockManagerMaster - BlockManagerMaster stopped
56052 [dispatcher-event-loop-4] INFO org.apache.spark.scheduler.OutputCommitCoordinator$OutputCommitCoordinatorEndpoint - OutputCommitCoordinator s
topped!
56067 [Thread-5] INFO org.apache.spark.SparkContext - Successfully stopped SparkContext
56634 [shutdown-hook-0] INFO org.apache.spark.util.ShutdownHookManager - Shutdown hook called
56635 [shutdown-hook-0] INFO org.apache.spark.util.ShutdownHookManager - Deleting directory /tmp/spark-bb33db8b-b69d-4f1a-8b8f-7dc52d961257
56640 [shutdown-hook-0] INFO org.apache.spark.util.ShutdownHookManager - Deleting directory /tmp/spark-25615cc8-3d66-4dc3-b0c9-0b16e92a02f2
56646 [shutdown-hook-0] INFO org.apache.spark.util.ShutdownHookManager - Deleting directory /tmp/spark-bb33db8b-b69d-4f1a-8b8f-7dc52d961257/pyspark
-6499d468-e319-41d0-b5ce-309f7c1d6e95
bash-5.0# hbase shell
2026-03-03 03:24:24,860 WARN [main] util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes wher
e applicable
HBase Shell
Use "help" to get list of supported commands.
Use "exit" to quit this interactive shell.
For Reference, please visit: http://hbase.apache.org/2.0/book.html#shell
Version 2.3.6, r7414579f2620fca6b75146c29ab2726fc4643ac9, Wed Jul 28 22:24:42 UTC 2021
Took 0.0012 seconds
hbase(main):001:0> scan 'car_sales'
ROW COLUMN+CELL
metrics1 column=details:r2, timestamp=2026-03-03T03:21:51.938, value=0.953208182243094
metrics1 column=details:rmse, timestamp=2026-03-03T03:21:51.938, value=2546.796411166555
wbsfv9c51ed593089 column=details:make, timestamp=2026-03-03T00:09:13.325, value=BMW
wbsfv9c51ed593089 column=details:model, timestamp=2026-03-03T00:09:13.355, value=M5
wbsfv9c51ed593089 column=details:state, timestamp=2026-03-03T00:09:16.662, value=ca
wbsfv9c51ed593089 column=details:year, timestamp=2026-03-03T00:09:13.286, value=2014
2 row(s)
Took 0.8903 seconds
hbase(main):002:0>
```

In this section, I used Linear Regression from Spark MLlib. Linear regression is appropriate for dataset because the target variable, sellingprice, is continuous and influenced by number features such as vehicle year, odometer reading and market value (MMR). The model was trained using a 70/30 train test split. The model was evaluated using Root Mean Squared Error and R2. Showing the strong predictive performance, show that the selected features are effective for predicting the vehicle selling prices.

Objective 6 – Spark Submit & Output:

Code in the terminal, master container.

Thrift server and Hbase are running.

```
meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$ docker-compose exec master bash
bash-5.0# cd /usr/program/source_dir
bash-5.0# spark-submit \
> --master yarn \
> --deploy-mode client \
> --name CarPricePrediction \
> car_prices_ml.py
```



```
meganpokal — meganpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189 — 1...
...pokal27@35.236.15.189 ... ..pokal27@35.236.15.189 ...oop-hive-spark-hbase ...spark-hbase — -bash ...kal27@35.236.15.189 +
[meganpokal27@dsc650bigdata:~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase$ docker-compose exec master bash
bash-5.0# cd /usr/program/source_dir
bash-5.0# spark-submit \
> --master yarn \
> --deploy-mode client \
> --name CarPricePrediction \
> car_prices_ml.py
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/program/spark/jars/slf4j-log4j12-1.7.30.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/program/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
0 [main] WARN org.apache.hadoop.util.NativeCodeLoader - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
1340 [Thread-5] INFO org.apache.spark.SparkContext - Running Spark version 3.0.0
1390 [Thread-5] INFO org.apache.spark.resource.ResourceUtils - =====
1392 [Thread-5] INFO org.apache.spark.resource.ResourceUtils - Resources for spark.driver:

1392 [Thread-5] INFO org.apache.spark.resource.ResourceUtils - =====
1393 [Thread-5] INFO org.apache.spark.SparkContext - Submitted application: CarPriceML
1468 [Thread-5] INFO org.apache.spark.SecurityManager - Changing view acls to: root
1468 [Thread-5] INFO org.apache.spark.SecurityManager - Changing modify acls to: root
1468 [Thread-5] INFO org.apache.spark.SecurityManager - Changing view acls groups to:
1468 [Thread-5] INFO org.apache.spark.SecurityManager - Changing modify acls groups to:
1469 [Thread-5] INFO org.apache.spark.SecurityManager - SecurityManager: authentication disabled; ui acls disabled; users with view permissions: Set(et(root); groups with view permissions: Set(); users with modify permissions: Set(root); groups with modify permissions: Set()
1823 [Thread-5] INFO org.apache.spark.util.Utils - Successfully started service 'sparkDriver' on port 7001.
1866 [Thread-5] INFO org.apache.spark.SparkEnv - Registering MapOutputTracker
1907 [Thread-5] INFO org.apache.spark.SparkEnv - Registering BlockManagerMaster
1933 [Thread-5] INFO org.apache.spark.storage.BlockManagerMasterEndpoint - Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
1933 [Thread-5] INFO org.apache.spark.storage.BlockManagerMasterEndpoint - BlockManagerMasterEndpoint up
1991 [Thread-5] INFO org.apache.spark.SparkEnv - Registering BlockManagerMasterHeartbeat
2008 [Thread-5] INFO org.apache.spark.storage.DiskBlockManager - Created local directory at /tmp/blockmgr-faa8b1cc-6f32-4d7f-8431-80cb8cc86070
2046 [Thread-5] INFO org.apache.spark.storage.memory.MemoryStore - MemoryStore started with capacity 366.3 MiB
2109 [Thread-5] INFO org.apache.spark.SparkEnv - Registering OutputCommitCoordinator
2260 [Thread-5] INFO org.sparkproject.jetty.util.log - Logging initialized @4245ms to org.sparkproject.jetty.util.log.Slf4jLog
```

```
meganpokal — meganpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh meganpokal27@35.236.15.189 — 1...
...pokal27@35.236.15.189 ... ..pokal27@35.236.15.189 ...oop-hive-spark-hbase ...spark-hbase — -bash ...kal27@35.236.15.189 +
45757 [dispatcher-BlockManagerMaster] INFO org.apache.spark.storage.BlockManagerInfo - Added broadcast_13_piece0 in memory on worker1:7002 (size: 4.1 KiB, free: 366.3 MiB)
46526 [task-result-getter-0] INFO org.apache.spark.scheduler.TaskSetManager - Finished task 1.0 in stage 7.0 (TID 13) in 826 ms on worker2 (executor 1) (1/2)
46528 [dag-scheduler-event-loop] INFO org.apache.spark.api.python.PythonAccumulatorV2 - Connected to AccumulatorServer at host: 127.0.0.1 port: 54475
46559 [task-result-getter-1] INFO org.apache.spark.scheduler.TaskSetManager - Finished task 0.0 in stage 7.0 (TID 12) in 863 ms on worker1 (executor 2) (2/2)
46559 [task-result-getter-1] INFO org.apache.spark.scheduler.cluster.YarnScheduler - Removed TaskSet 7.0, whose tasks have all completed, from pool
46561 [dag-scheduler-event-loop] INFO org.apache.spark.scheduler.DAGScheduler - ResultStage 7 (foreachPartition at /usr/program/source_dir/car_prices_ml.py:50) finished in 0.872 s
46561 [dag-scheduler-event-loop] INFO org.apache.spark.scheduler.DAGScheduler - Job 5 is finished. Cancelling potential speculative or zombie tasks for this job
46561 [dag-scheduler-event-loop] INFO org.apache.spark.scheduler.cluster.YarnScheduler - Killing all running tasks in stage 7: Stage finished
46562 [Thread-5] INFO org.apache.spark.scheduler.DAGScheduler - Job 5 finished: foreachPartition at /usr/program/source_dir/car_prices_ml.py:50, took 0.880136 s
46609 [Thread-5] INFO org.sparkproject.jetty.server.AbstractConnector - Stopped Spark@f80be3a(HTTP/1.1,[http://1.1]{172.28.1.1:4040})
46612 [Thread-5] INFO org.apache.spark.ui.SparkUI - Stopped Spark web UI at http://localhost:4040
46623 [YARN application state monitor] INFO org.apache.spark.scheduler.cluster.YarnClientSchedulerBackend - Interrupting monitor thread
46655 [Thread-5] INFO org.apache.spark.scheduler.cluster.YarnClientSchedulerBackend - Shutting down all executors
46656 [dispatcher-CoarseGrainedScheduler] INFO org.apache.spark.scheduler.cluster.YarnSchedulerBackend$YarnDriverEndpoint - Asking each executor to shut down
46677 [Thread-5] INFO org.apache.spark.scheduler.cluster.YarnClientSchedulerBackend - YARN client scheduler backend Stopped
47135 [dispatcher-event-loop-7] INFO org.apache.spark.MapOutputTrackerMasterEndpoint - MapOutputTrackerMasterEndpoint stopped!
47155 [Thread-5] INFO org.apache.spark.storage.memory.MemoryStore - MemoryStore cleared
47156 [Thread-5] INFO org.apache.spark.storage.BlockManager - BlockManager stopped
47162 [Thread-5] INFO org.apache.spark.storage.BlockManagerMaster - BlockManagerMaster stopped
47170 [dispatcher-event-loop-4] INFO org.apache.spark.scheduler.OutputCommitCoordinator$OutputCommitCoordinatorEndpoint - OutputCommitCoordinator stopped!
47187 [Thread-5] INFO org.apache.spark.SparkContext - Successfully stopped SparkContext
47665 [shutdown-hook-0] INFO org.apache.spark.util.ShutdownHookManager - Shutdown hook called
47667 [shutdown-hook-0] INFO org.apache.spark.util.ShutdownHookManager - Deleting directory /tmp/spark-3fda08d5-036c-4237-9a80-b56a640cad7c
47674 [shutdown-hook-0] INFO org.apache.spark.util.ShutdownHookManager - Deleting directory /tmp/spark-3fda08d5-036c-4237-9a80-b56a640cad7c/pyspark-1ea12a69-634a-4d68-b85c-9ed31bf73368
47679 [shutdown-hook-0] INFO org.apache.spark.util.ShutdownHookManager - Deleting directory /tmp/spark-7d842ad7-d64c-4a2b-9f0b-90f638b893d6
bash-5.0#
```

Objective 7 – HBase Scan Verification:

In the master container.

hbase shell

```
meganpokal — megalpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh megalpokal27@35.236.15.189 — 1...
...pokal27@35.236.15.189 ... ..pokal27@35.236.15.189 ...oop-hive-spark-hbase ...spark-hbase — -bash ...kal27@35.236.15.189 +
47679 [shutdown-hook-0] INFO org.apache.spark.util.ShutdownHookManager - Deleting directory /tmp/spark-7d842ad7-d64c-4a2b-9f0b-90f638b893d6
bash-5.0# hbase shell
2026-03-03 03:35:26,219 WARN [main] util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
HBase Shell
Use "help" to get list of supported commands.
Use "exit" to quit this interactive shell.
For Reference, please visit: http://hbase.apache.org/2.0/book.html#shell
Version 2.3.6, r7414579f2620fca6b75146c29ab2726fc4643ac9, Wed Jul 28 22:24:42 UTC 2021
Took 0.0024 seconds
hbase(main):001:0>
```

Scan 'car_sales'

```
meganpokal — megalpokal27@dsc650bigdata: ~/dsc650-infra/bellevue-bigdata/hadoop-hive-spark-hbase — ssh megalpokal27@35.236.15.189 — 1...
...pokal27@35.236.15.189 ... ..pokal27@35.236.15.189 ...oop-hive-spark-hbase ...spark-hbase — -bash ...kal27@35.236.15.189 +
hbase(main):001:0> scan 'car_sales'
ROW COLUMN+CELL
metrics1 column=details:r2, timestamp=2026-03-03T03:31:53.256, value=0.8833298994832645
metrics1 column=details:rmse, timestamp=2026-03-03T03:31:53.295, value=4321.798540172366
wbsfv9c51ed593089 column=details:make, timestamp=2026-03-03T00:09:13.325, value=BMW
wbsfv9c51ed593089 column=details:model, timestamp=2026-03-03T00:09:13.355, value=M5
wbsfv9c51ed593089 column=details:state, timestamp=2026-03-03T00:09:16.662, value=ca
wbsfv9c51ed593089 column=details:year, timestamp=2026-03-03T00:09:13.286, value=2014
2 row(s)
Took 0.6987 seconds
hbase(main):002:0>
```

After the Spark ML job had completed, the evaluated metrics were written into the HBase table car_sales. The row key metrics1 contains two columns, details:rmse and details:r2. RSME shows the average prediction error of the model, while R2 shows how well the model explains the variance in the dataset. The presence of these metrics in HBase confirms that the pipeline did work successfully, executing a end to end pipeline. As the data was put into HDFS and processed with Hive and Spark, while the machine learning results were stored in HBase.

Issues Encountered

There were a few issues that occurred while trying to do this final project. In the beginning, it was difficult to pick the right dataset and then having to size it down correctly to make it work for this project, with limited space in the VM instance. One issue was the missing Python dependencies such as PySpark and HappyBase, when attempting to run the ML script locally. This was resolved by installing the required packages using pip. Another issue was when the Spark script couldn't locate the ML file inside the Docker container. This was resolved by correctly copying the file into the container's source directory.

Conclusion

In this project, it will show the end to end big data pipeline using Nifi, HDFS, Hive, Spark, and HBase. Data was ingested from GitHub using Nifi and stored in HDFS. Hive provided structured access to the dataset, and Spark MLLib was used to train a linear regression model predicting vehicle selling prices. The model evaluation metrics were written into HBase and then verified with Hbase shell. This project shows how the different / multiple technologies in the Hadoop

system can work together and support data processing and machine learning workflows. Thank you!